Lekh Nayak

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Education

Sardar Patel Institute Of Technology **Aug 2023 – Present**
B.Tech in Computer Science Engineering *CGPA: 8.05 /10*

IIT Patna **Mar 2025 – Present**
Minor in Business Analytics *CGPA: 9.5/10*

Experience

GPU Access Portal **Dec 2025 – Feb 2026**
System Engineering Intern *Sardar Patel Institute of Technology*

- Built a scalable HPC cluster using **Slurm**, **NFS**, and **SSSD/LDAP**, while integrating **Prometheus** and **Grafana** for real-time monitoring of GPU utilization, job queues, and node health.
- Deployed and configured **Open OnDemand** as a web-based HPC portal via **direct SSH root access**, enabling **browser-native access** to **Jupyter Notebook, VS Code Server, and RStudio** for students without any CLI setup.

Cactus Life Sciences **Jun 2026 – Present**
AI Automation Intern *Remote*

- Building a **SaaS platform** that unifies **LLM-powered text, infographic, and presentation generation** into a single content creation workflow.
- Developing **multi-agent AI workflows**, **LLM integrations**, and automation pipelines to accelerate research, drafting, and content production.

Projects

InterviewMate | *React, Node.js, Express, MongoDB, LangChain, Pinecone, OpenAI API* **Sept 2025 - Dec 2025**

- Architected a **personalized interview preparation platform** with a **RAG pipeline** using LangChain and Pinecone, achieving **˜85% retrieval accuracy** across a knowledge base of **500+ domain-specific Q&A pairs** for context-aware mock question generation.
- Designed **session-based storage and analytics** in MongoDB, tracking feedback scores and performance trends across **50+ simulated interview sessions**, enabling users to identify weak areas across **5+ topic categories**.
- Built an **intelligent QnA interface** integrating OpenAI's GPT model for **adaptive difficulty scaling** and follow-up explanations, generating **10+ contextual follow-up questions** per session based on user responses.

Collaborative VS Code | *React, Node.js, Express, WebSocket, Monaco Editor, Docker, PostgreSQL* **Jan 2026 - Mar 2026**

- Engineered a **real-time collaborative code editor** stress-tested with **30+ concurrent users**, maintaining **synchronized cursors, terminals, and file states** with under **100ms sync latency**.
- Integrated **Monaco Editor** with syntax highlighting and intelligent suggestions across **6+ languages**, executing code in **Docker-isolated sandboxed environments** for safe multi-user runs.
- Designed a **WebSocket-driven event system** handling real-time collaboration, code execution broadcasts, and **state consistency** across all connected clients with zero data loss on reconnection.

CtrlFixIt | *Next.js, TypeScript, PostgreSQL, Redis, Docker, WebSockets, Kafka* **Jan 2026 - Feb 2026**

- Built a **production-grade competitive coding platform** deployed live for **300+ participants**, handling **150+ concurrent sandboxed code execution jobs** via a Piston-based Docker engine clustered with Kafka and tunneled through ngrok.
- Engineered a **real-time event system** using WebSockets for live scoreboards, instant code execution feedback, and **server-authoritative timing** to ensure fairness across all connected clients.
- Optimized for **vertical scalability** using Redis-backed queuing and Prisma ORM with PostgreSQL, supporting multi-language execution (Python, C++, Java, Go, Rust, JavaScript) with **1GB isolated temp memory** per job.

Technical & Soft Skills

Programming Languages: Java, Python, JavaScript, SQL
Libraries/Frameworks: React, Express.js, Node.js, PyTorch, Scikit-learn, NLTK, BERT
Developer Tools: Git, Docker
Soft Skills: Team Management, Team Collaboration, Public Speaking

Position of Responsibility

Computer Society of India **Aug 2025 – May 2026**
Head of Events *Sardar Patel Institute of Technology*

- Led chapter operations by managing a 60+ member team, driving strategic initiatives, and organizing technical events for 350+ participants through effective stakeholder coordination and end-to-end execution.

Extra Curricular & Achievements

- **1st Place:** La Casa De Code - National Level Hackathon (Dayananda Sagar University), 2x Abhimat - Lok Sabha Style Debate, CSI S.P.I.T ML Competition (Sardar Patel Institute of Technology, Mumbai)
- **2nd Place:** EFOS Virtual Hackathon - EFOS, Runner up Best Delegate - AIPPM (mUNIFY S.P.I.T. Mumbai), MergeMania - Mergers & Acquisition (E-Summit 2025 S.P.I.T. Mumbai), Dreams and Deals - Pitching Contest, Pitch Hunt - Elevator Pitch Challenge, Annual Oxford Style Debate (S.P.C.E. Mumbai)
- **Finalist:** HERE Geospatial Hackathon (Pan India Hiring Hack), Minithon (IIIT Bhubaneswar – top 9/60+), Airavat AI Hackathon (SPIT Mumbai 2024), Inspiriobiz 5.0 (VESIM Mumbai – top 10/300+ teams), Vishleshan – Canfem Case Study (TISS Mumbai – top 5/550+ teams), BizQuest (E-Summit 2024 SPIT Mumbai)